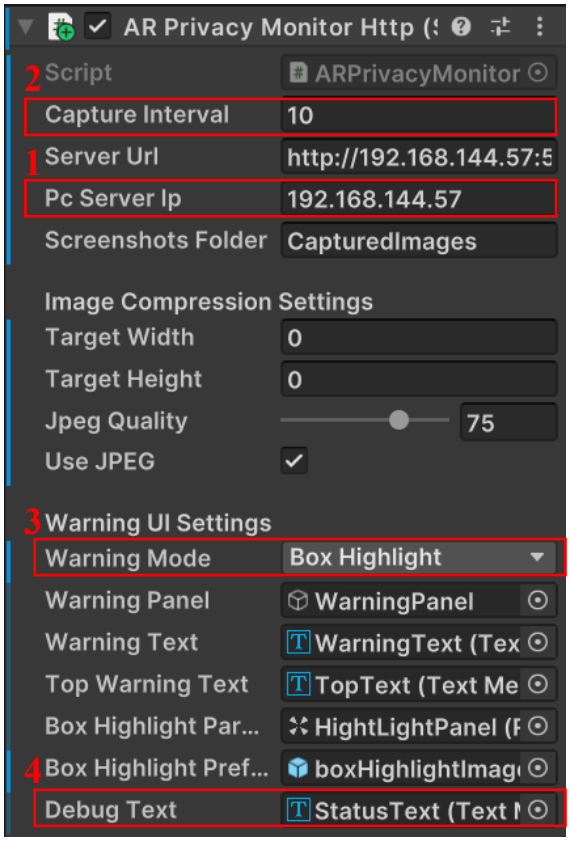
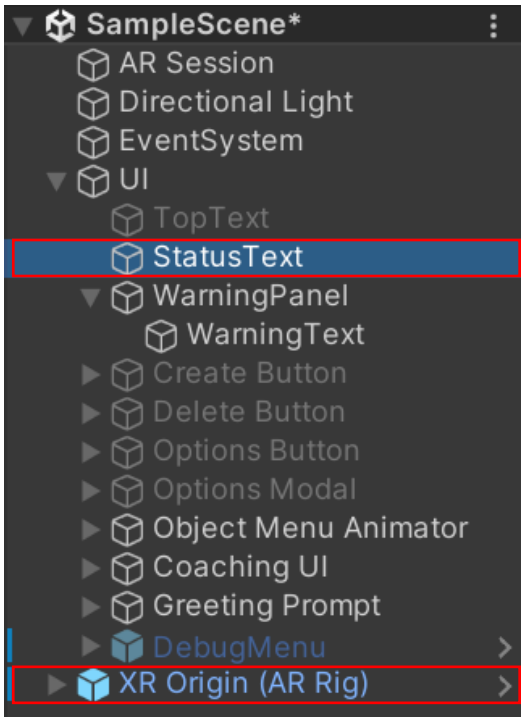
Construction Guide for AR Apps with Different Warning Modes

Prerequisite:

The mobile device and the computer server must be connected to the same Wi-Fi network (preferably not a public Wi-Fi).

Step 1: Modify parameters in Unity



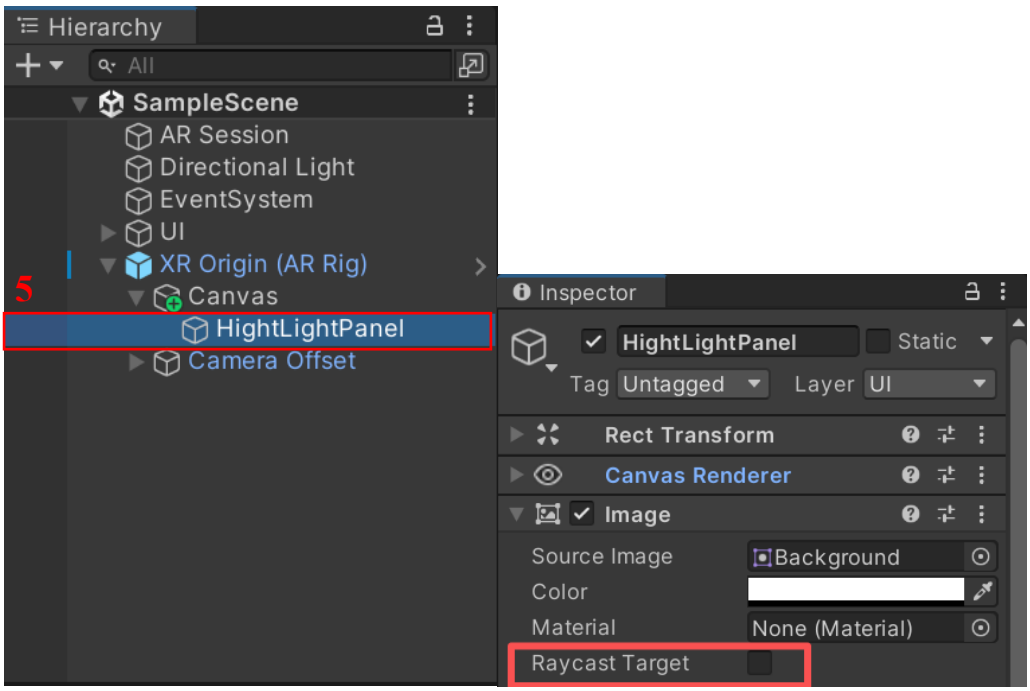


The setting of parameters is at the *ARPrivacyMonitorHttp.cs* script, which is the component of *XR Origin(AR Rig)* GameObject.

- **1. Set IP address:** Change it to the IP address of the Wi-Fi being used.

  Only the *Pc Server Ip* parameter needs to be set.
- **2. Set trigger interval:** The interval can be configured between 20s and 30s.
- **3. Set warning mode:** For different app versions corresponding to different warning modes, select the three related variables.

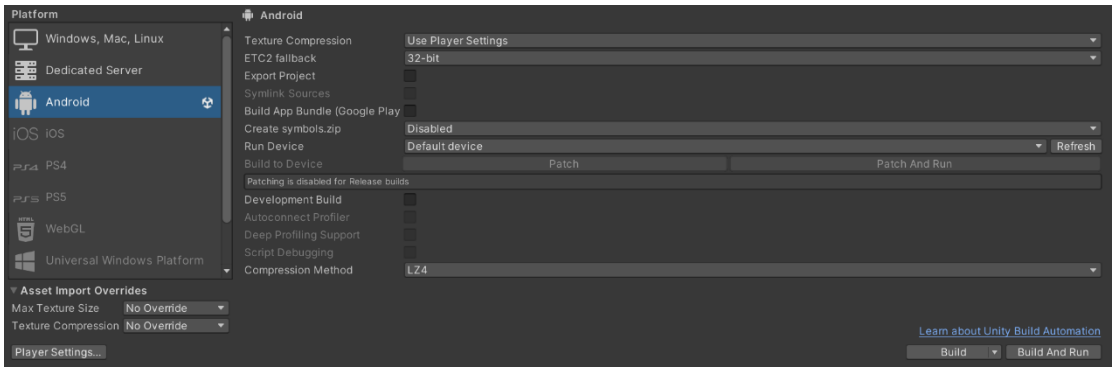
  Three modes: *Center Panel/Top Text/Box Highlight*
- **4. Debug option:** If you want to monitor the execution process on the mobile device to test whether the entire pipeline runs successfully, check the option *StatusText*. For formal experiments, uncheck this option.
- **5.** Uncheck the box of the *'Raycast Target'*: In order to enable the two tasks of AR privacy information detection and virtual object placement on the AR plane to run at the same time, we need to uncheck the box of the *'Raycast Target'* in the component of *'HightLightPenal'*:



Step 2: Build the AR app

- Connect the phone to the computer via USB cable.
- Enable developer mode and USB debugging on the phone.

- Open *Unity* → *File* → *Build Settings* → *Run Device*, find the connected mobile device and select it (minimum Android version: 29).



- Click *Build And Run* to start the build process.

Step 3: Start the edge server

- Open the file *ARtest/Assets/MobileARTemplateAssets/Scripts/privacy_http_server.py* and set the variable *EAST_model_path* to the *model/frozen_east_text_detection.pb* file in the compressed package.

```python
import numpy as np
import pytesseract as ts


app = Flask(__name__)
@app.route('/check_privacy', methods=['POST'])
def check_privacy():
    print("check_privacy")
    gpt = GPTAzure("gpt-4o-mini")
    print('gpt loaded!')
    EAST_model_path="E:/Study/model/frozen_east_text_detection.pb"
```

- Run this code file.
- **Test connection:** Open the webpage *http://your_ip:5000/check_privacy* on the phone. The phone may return error code=405, but the key point is to check whether the Python process outputs a response. If it does, the connection is successful.

Step 4: Start the experiment

- Run the AR app on the mobile device. At the beginning, the software needs a short initialization period.
- After initialization, it will capture an image every set interval, upload it to the edge server for analysis, and then return feedback.
- If a privacy-related object is detected by the camera but no warning feedback appears, enable *StatusText* and rebuild the app to display step outputs on the mobile device. This helps identify where the problem occurs.